Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 50 Question Id : 695278425 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In a double slit experiment, when one of the slits is covered by a transparent mica sheet of refractive index 1.56, the central fringe shifts to the position of 7th bright fringe, obtained with both slits uncovered. If the light source wavelength is 450 nm, the thickness of mica sheet is $\alpha \times 10^{-9}$ m. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	69527835
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527835
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 695278426 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The correct order of total number of atoms present in

(A) 2 moles of cyclohexane

(B) 684 g of sucrose

(C) 90.8 L of dihydrogen at STP

is:

Options :

6952781446. $C > A > B$

6952781447. $C > B > A$

6952781448. $B > C > A$

6952781449. $B > A > C$

Question Number : 52 Question Id : 695278427 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The species having identical radii according to the Bohr's theory are:

A. H (first orbit)
B. He^{+} (first orbit)
C. He^{+} (Second orbit)
D. Li^{2+} (first orbit)
E. Be^{3+} (Second orbit)

Choose the correct answer from the options given below:

Options :

6952781450. A and C Only

6952781451. A and E Only

6952781452. B and E Only

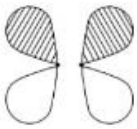
6952781453. C and D Only

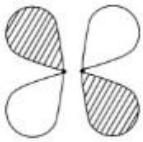
Question Number : 53 Question Id : 695278428 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

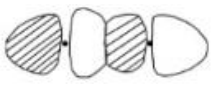
Which of the following pictorial diagram most correctly represents the π^* (π - antibonding) molecular orbital between two atoms if the internuclear axis is taken to be in the z-direction ($\xrightarrow{\text{z-axis}}$) ?

Options :

6952781454. 

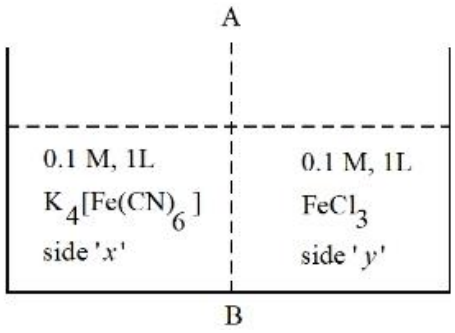
6952781455. 

6952781456. 

6952781457. 

Question Number : 54 Question Id : 695278429 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

At 27 °C, 0.1 M, 1 L $K_4[Fe(CN)_6]$ aqueous solution and 0.1 M, 1 L $FeCl_3$ aqueous solution are placed in a container separated by a semi permeable membrane AB. Assume complete dissociation of both the solutes. Which of the following statement is ***correct***?



Options :

6952781458. Blue color is formed on both sides.

6952781459. Ionic solutes in aqueous solution can pass through semi-permeable membrane.

6952781460. Solution on side 'y' is hypotonic.

6952781461. To cause the reverse flow of solvent during osmosis, external pressure (any value) should be applied to side '*x*'.

Question Number : 55 Question Id : 695278430 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the above acetic acid and 14.2 mL of 0.1 M NaOH solution. What is the pH of the solution (X)? (pK_a value of acetic acid is 4.75).

Options :

6952781462. 7.0

6952781463. 4.75

6952781464. 3.5

6952781465. 4.82

Question Number : 56 Question Id : 695278431 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I **Reaction**		**List-II** **Mechanism**
A.	Williamson Synthesis	I.	Electrophilic addition
B.	Friedel Craft Reaction	II.	Free radical substitution
C.	Bromination of vinyl benzene	III.	Nucleophilic substitution
D.	Chlorination of toluene in light	IV.	Electrophilic substitution

Choose the ***correct*** answer from the options given below:

Options :

6952781466. A-III, B-I, C-II, D-IV

6952781467. A-III, B-IV, C-II, D-I

6952781468. A-III, B-IV, C-I, D-II

6952781469. A-I, B-III, C-IV, D-II

Question Number : 57 Question Id : 695278432 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The 1st ionization enthalpy for Mg is +737 kJ/mol. The most probable estimated value of the 2nd ionization enthalpy of Mg is _____.

Options :

6952781470. −906 kJ/mol

6952781471. −856 kJ/mol

6952781472. +1450 kJ/mol

6952781473. +590 kJ/mol

Question Number : 58 Question Id : 695278433 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The electronegativity of a group 13 element 'E' is same as that of Ge (on Pauling scale and upto one decimal point). The **CORRECT** statements about E^{3+} are

A. It can act as a reducing agent.
B. It can act as an oxidizing agent.
C. E^{3+} is more stable than E^{+}.
D. The standard electrode potential value for E^{3+}/E is positive.

Choose the correct answer from the options given below:

Options :

6952781474. A and C Only

6952781475. B and C Only

6952781476. B and D Only

6952781477. A and D Only

Question Number : 59 Question Id : 695278434 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Pairs of elements with the same number of electrons in their respective 4f orbital are

[Atomic number. Eu-63, Gd-64, Dy-66, Ho-67, Tm-69, Yb-70, Lu-71, Hf-72]

A. (Eu and Gd)
B. (Dy and Ho)
C. (Yb and Hf)
D. (Lu and Tm)

Choose the correct answer from the options given below:

Options :

6952781478. B and C Only

6952781479. A and B Only

6952781480. A and D Only

6952781481. A and C Only

Question Number : 60 Question Id : 695278435 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the metal complexes $[Ni(en)_3]^{2+}$ (A), $[NiCl_4]^{2-}$ (B) and $[Ni(NH_3)_6]^{2+}$(C). Choose the **CORRECT** option by considering the number of unpaired electrons present in (A), (B) and (C) respectively and the order of frequency of absorption.

Options :

6952781482. 2, 2, 2 and (A) > (C) > (B)

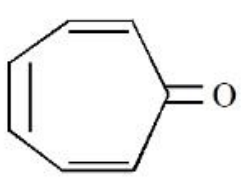
6952781483. 0, 2, 0 and (A) > (C) > (B)

6952781484. 2, 2, 0 and (B) > (C) > (A)

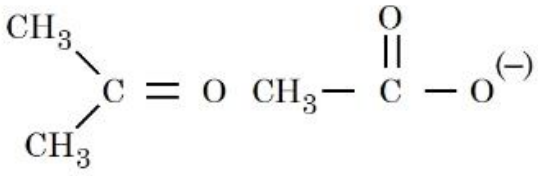
6952781485. 2, 2, 2 and (C) > (A) > (B)

Question Number : 61 Question Id : 695278436 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following molecules/species:



(*x*)



(y)

(z)

The correct order of carbon – oxygen double bond length is :

Options :

6952781486. *x* > y > z

6952781487. $y > z > x$

6952781488. $z > x > y$

6952781489. $x > z > y$

Question Number : 62 Question Id : 695278437 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider $|x|$ is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with $|x|$ number of unpaired electrons from the following are:

A. Sc^{3+}
B. Zn^{2+}
C. V^{2+}
D. Fe^{2+}
E. Co^{2+}

Choose the correct answer from the options given below:

Options :

6952781490. A and B Only

6952781491. C, D and E Only

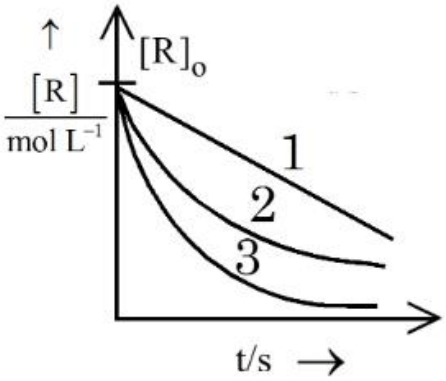
6952781492. C and E Only

6952781493. B and E Only

Question Number : 63 Question Id : 695278438 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the given graph showing variation of reactant concentration with time.

Three different reactions were started with identical initial concentration of reactants. Which of the following statement is correct?



Options :

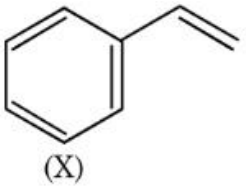
6952781494. The order of all the three reactions is same.

6952781495. The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

6952781496. The SI unit of rate constant of reaction 1 is s^{-1}.

6952781497. Thermal decomposition of HI on gold surface is an example of reaction 2.

Question Number : 64 Question Id : 695278439 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical



(X)

(i) Br_2 / $CHCl_3$
(ii) $NaNH_2$ excess
(iii) CH_3I
(iv) H_2, Na/NH_3 (l)

→ Major Product (Y)

Compound (X) is subjected to the sequence of reactions as shown above. Molar mass of the major product (Y) formed is _____ g mol^{-1}.

(Given molar mass in g mol^{-1} C:12, H: 1, O: 16)

Options :

6952781498. 90

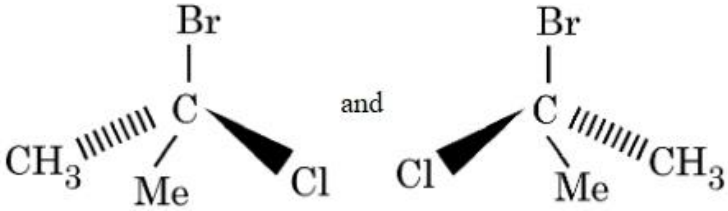
6952781499. 118

6952781500. 160

6952781501. 125

Question Number : 65 Question Id : 695278440 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The following structures are



Options :

6952781502. enantiomers.

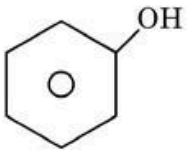
6952781503. identical molecules.

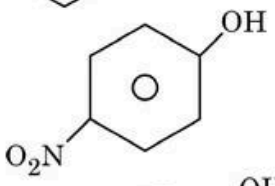
6952781504. diastereomers.

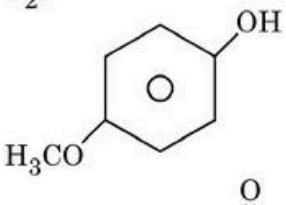
6952781505. meso compounds.

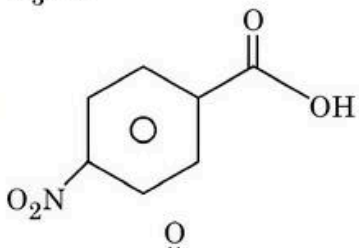
Question Number : 66 Question Id : 695278441 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

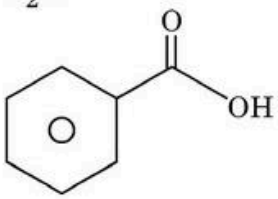
The descending order of acidity among the following compounds is:

A. 

B. 

C. 

D. 

E. 

Choose the correct answer from the options given below:

Options :

6952781506. $B > D > E > A > C$

6952781507. $D > B > E > A > C$

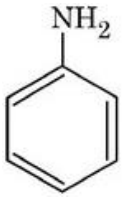
6952781508. $C > A > B > D > E$

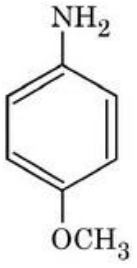
6952781509. $D > E > B > A > C$

Question Number : 67 Question Id : 695278442 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The strongest conjugate acid will result from:

Options :

6952781510. 

6952781511. 

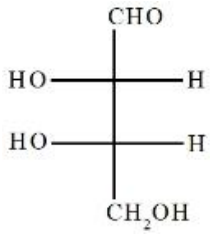
6952781512. 

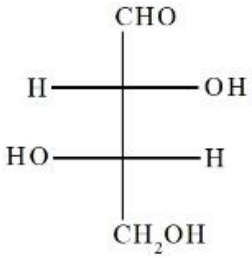
6952781513. 

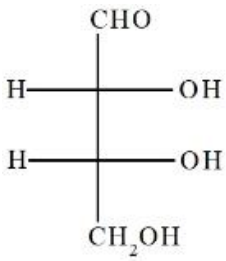
Question Number : 68 Question Id : 695278443 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

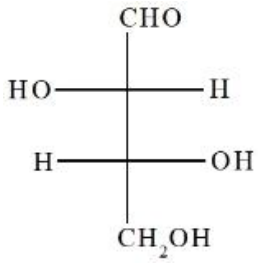
A D-aldotetrose on oxidation with concentrated HNO_3 resulted in optically inactive dicarboxylic acid. The structure of the D-aldotetrose is:

Options :

6952781514. 

6952781515. 

6952781516. 



6952781517.

Question Number : 69 Question Id : 695278444 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Among Fe^{3+}, Pb^{2+}, Cu^{2+} and Mn^{2+}, identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

Options :

6952781518. $+3$

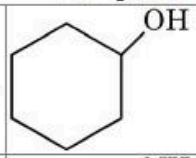
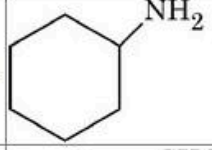
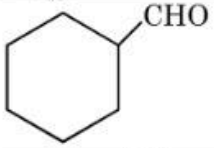
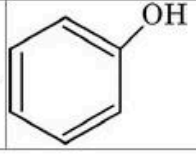
6952781519. $+4$

6952781520. $+2$

6952781521. $+7$

Question Number : 70 Question Id : 695278445 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I Compound		**List-II Test**
A.		I.	Hinsberg's reagent test
B.		II.	Phthalein dye test
C.		III.	Lucas test
D.		IV.	Tollen's test

Choose the ***correct*** answer from the options given below:

Options :

6952781522. A-III, B-I, C-IV, D-II

6952781523. A-III, B-IV, C-I, D-II

6952781524. A-I, B-III, C-II, D-IV

6952781525. A-I, B-II, C-III, D-IV

Chemistry Section B

Section Id :	69527836
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527836
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 695278446 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If 3.365g of ethanol (*l*) is burnt completely in a bomb calorimeter at 298.15 K, the heat produced is 99.472 kJ. The $|\Delta H_f^\circ|$ of ethanol at 298.15 K is

_____ $\times 10^2$ kJ mol^{-1}. (Nearest integer)

Given: Standard enthalpy for combustion of graphite = -393.5 kJ mol^{-1}

Standard enthalpy of formation of water (l) = -285.8 kJ mol^{-1}

Molar mass in g mol^{-1} of C, H, O are 12 , 1 and 16 respectively

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 695278447 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For the following reaction at 50 °C and at 2 atm pressure,

$2N_2O_5(g) \rightleftharpoons 2N_2O_4(g) + O_2(g)$

N_2O_5 is 50% dissociated.

The magnitude of standard free energy change at this temperature is *x*.

x = _____ J mol^{-1} [Nearest integer].

Given : R = 8.314 J mol^{-1} K^{-1}, log 2 = 0.30, log 3 = 0.48, ln10 = 2.303,
°C + 273 = K

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Possible Answers :

1

Question Number : 73 Question Id : 695278448 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

An electrochemical cell, consist of the following two redox couples, M^{x+} (aq)/M(s) $[E^{\Theta}_{red} = +0.15\,V]$ and Fe^{3+}(aq)/Fe(s) $[E^{\Theta}_{red} = -0.036\,V]$. The cell EMF ($E_{cell}$) is recorded to be 0.2057 V. If the reaction quotient of the electrochemical reaction is found to be 10^{-2}, then the value of x is _____.(Nearest integer)

[Given : M is a p-block metal and $\frac{2.303RT}{F} = 0.059$ V]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 74 Question Id : 695278449 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For a first order reaction A → B

t/min	[A]/M
0	0.6500
x	0.0650
20	0.00065

x = _____ min. (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 75 Question Id : 695278450 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In sulphur estimation, 2.0×10^{-3} mol of an organic compound (X) (molar mass 76 g mol^{-1}) gave 0.4813 g of barium sulphate (molar mass 233 g mol^{-1}). The percentage of sulphur in the compound (X) is _____ $\times 10^{-1}$ % (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1